

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

## Project: Online Internship & Placement Preparation Portal

**Version:** 1.0

**Authors:**

**Date:**

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## Revision History

| Version | Date       | Author | Change Summary       | Approval |
|---------|------------|--------|----------------------|----------|
| 1.0     | 05-09-2026 |        | Initial SRS document | Pending  |

## Approvals

| Role               | Name | Signature / Email | Date |
|--------------------|------|-------------------|------|
| Course Coordinator |      |                   |      |
| Project Guide      |      |                   |      |
| Team Members       |      |                   |      |

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## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the **Online Internship & Placement Preparation Portal**.

The system is designed to provide students with a single platform to search and apply for internships, attend aptitude tests, monitor preparation progress, and view performance analytics.

The system also provides facilities for recruiters to post internship opportunities and manage applications, while administrators can manage users, internships, tests, and system activities.

This document describes the functional requirements, non-functional requirements, security requirements, interfaces, system features, and acceptance criteria of the proposed system.

### 1.2 Scope

The Online Internship & Placement Preparation Portal provides an integrated platform for internship discovery and placement preparation.

The system includes:

- Student registration and login
- Student profile management
- Internship listing and searching
- Internship application
- Application status tracking

- Aptitude tests
- Automatic test evaluation
- Test history and performance tracking
- Placement preparation progress tracking
- Performance analytics
- Recruiter registration and login
- Internship posting and management
- Candidate application management
- Candidate shortlisting and rejection
- Administrator management of users, internships, tests and questions

The system does not directly conduct company interviews, provide employment guarantees, or perform final recruitment decisions outside the platform.

## 1.3 Audience

This document is intended for:

- Students
- Project developers
- Project guide
- Faculty members
- Recruiters
- System administrators
- Testers and evaluators

## 1.4 Definitions and Acronyms

| Term          | Definition                                                                |
|---------------|---------------------------------------------------------------------------|
| SRS           | Software Requirements Specification                                       |
| UI            | User Interface                                                            |
| API           | Application programming Interface                                         |
| MERN          | MongoDB, Express.js, React.js and Node.js                                 |
| FR            | Functional Requirement                                                    |
| NFR           | Non-Functional Requirement                                                |
| RTM           | Requirements Traceability Matrix                                          |
| OTP           | One-Time Password                                                         |
| RBAC          | Role-Based Access Control                                                 |
| Student       | User who searches and applies for internships and takes preparation tests |
| Recruiter     | User who posts internships and manages applications                       |
| Admin         | User responsible for managing the overall system                          |
| Aptitude Test | Online test used to evaluate a student's placement preparation            |
| Analytics     | Analysis of student test performance and preparation progress             |

## 2. Overall Description

### 2.1 Product Perspective

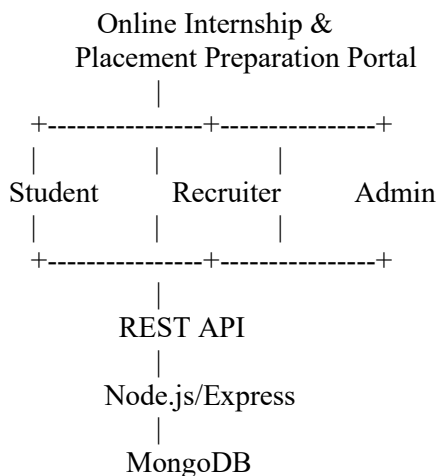
The Online Internship & Placement Preparation Portal is a web-based application developed using the MERN stack.

The system consists of:

- React.js frontend
- Node.js and Express.js backend
- MongoDB database
- REST APIs for communication between frontend and backend
- Authentication and role-based access control
- Analytics module for student performance

The system connects students, recruiters and administrators through a common platform.

#### Basic system architecture



### 2.2 Major Product Functions

The major functions of the system are:

#### Student Functions

- Student registration and login
- Profile management
- Internship search
- Internship filtering
- Internship application

- Application status tracking
- Aptitude test participation
- Automatic test evaluation
- Test history
- Preparation progress tracking
- Performance analytics

### Recruiter Functions

- Recruiter registration and login
- Recruiter profile management
- Internship creation
- Internship editing and deletion
- Viewing applications
- Candidate shortlisting
- Candidate rejection
- Application status updates

### Admin Functions

- User management
- Internship management
- Aptitude test management
- Question management
- Monitoring applications
- Viewing system-level analytics

## 2.3 User Roles and Characteristics

### Student

Students use the system to find internship opportunities and prepare for placement examinations.

Expected characteristics:

- Basic computer knowledge
- Can register and maintain a profile
- Can search and apply for internships
- Can take aptitude tests
- Can monitor preparation progress

## Recruiter

Recruiters represent companies and use the system to publish internship opportunities and manage applications.

Expected characteristics:

- Basic knowledge of recruitment activities
- Can create internship listings
- Can view candidate applications
- Can shortlist or reject candidates

## Administrator

The administrator manages and monitors the overall platform.

Expected characteristics:

- Knowledge of system administration
- Can manage users and content
- Can manage aptitude tests and questions
- Can monitor platform activities

## 2.4 Operating Environment

The system shall operate in:

### Client Side

- Desktop computers
- Laptops
- Tablets
- Smartphones
- Modern web browsers such as Chrome, Edge, Firefox and Safari

### Server Side

- Node.js runtime environment
- Express.js server
- MongoDB database
- Internet connection

## Development Environment

- Visual Studio Code
- Node.js
- MongoDB
- Git/GitHub
- React.js

## 2.5 Constraints

The system has the following constraints:

- Internet connection is required for normal operation.
- Users must provide valid registration information.
- Only authenticated users can access protected functions.
- Students can apply only for available internships.
- Recruiters can manage only their own internship listings.
- Administrators have higher-level management privileges.
- The system depends on the availability of the database and server.
- The system should follow secure coding practices for handling user information.

## 3. External Interface Requirements

### 3.1 User Interface

The system shall provide a simple and responsive web interface.

#### Student Interface

The student dashboard shall provide:

- Profile
- Internship listings
- Search and filter
- Applied internships
- Aptitude tests
- Test results
- Preparation progress
- Performance analytics



## Recruiter Interface

The recruiter dashboard shall provide:

- Recruiter profile
- Create internship
- Manage internships
- View applications
- Candidate status management

## Admin Interface

The administrator dashboard shall provide:

- User management
- Internship management
- Test management
- Question management
- System analytics

## 3.2 Software Interfaces

The system shall communicate between the frontend and backend using REST APIs.

### Frontend

- React.js
- HTML
- CSS
- JavaScript

### Backend

- Node.js
- Express.js

### Database

- MongoDB

## API Communication

React.js



HTTP/HTTPS Requests



Express.js / Node.js



Database Queries



MongoDB

## 3.3 Communication Interfaces

- The system shall use HTTP/HTTPS for client-server communication.
- HTTPS should be used for secure deployment.
- REST APIs shall be used for communication between frontend and backend.
- JSON shall be used for data exchange between frontend and backend.
- Appropriate error responses shall be returned when requests fail.

## 4. System Features

The following requirements define the major functional features of the system.

At least **15 functional requirements** are required. This SRS contains **18 functional requirements**.

## 4.1 User Registration and Authentication

### Description

The system shall provide secure registration and login functionality for students and recruiters.

| Req ID    | Requirement                                                                            | Type       | Priority | Source / Stakeholder | Acceptance / Criteria / Test Case                                            | Dependencies          |
|-----------|----------------------------------------------------------------------------------------|------------|----------|----------------------|------------------------------------------------------------------------------|-----------------------|
| IPR-F-001 | The system shall allow students to register using valid personal and academic details. | Functional | High     | Student              | Valid registration creates a student account.<br>TC-Auth-01                  | Database              |
| IPR-F-002 | The system shall authenticate registered users using valid login credentials.          | Functional | High     | Student/Recruiter    | Valid credentials provide access to the appropriate dashboard.<br>TC-Auth-02 | Authentication module |
| IPR-F-003 | The system shall provide role-based access for Student, Recruiter and Admin users.     | Functional | High     | Admin                | Users can access only their authorized modules.<br>TC-Auth-03                | Authentication        |

## 4.2 Student Profile Management

### Description

Students shall be able to create and maintain their placement profile.

| Req ID    | Requirement                                                                                  | Type       | Priority | Source / Stakeholder | Acceptance / Criteria / Test Case                                       | Dependencies    |
|-----------|----------------------------------------------------------------------------------------------|------------|----------|----------------------|-------------------------------------------------------------------------|-----------------|
| IPR-F-004 | The system shall allow students to create and update their profile information.              | Functional | High     | Student              | Updated information is displayed after saving.<br>Test: TC-PROF-01      | Student Account |
| IPR-F-005 | The system shall allow students to maintain academic details, skills and resume information. | Functional | Medium   | Student              | Valid profile information can be added and updated.<br>Test: TC-PROF-02 | Student Profile |

## 4.3 Internship Search and Application

### Description

Students shall be able to discover suitable internship opportunities and submit applications.

| Req ID    | Requirement                                                                            | Type       | Priority | Source / Stakeholder | Acceptance / Criteria / Test Case                                    | Dependencies         |
|-----------|----------------------------------------------------------------------------------------|------------|----------|----------------------|----------------------------------------------------------------------|----------------------|
| IPR-F-006 | The system shall display available internship opportunities to students.               | Functional | High     | Student              | Available internships are displayed correctly.<br>Test: TC-INT-01    | Internship database  |
| IPR-F-007 | The system shall allow students to search and filter internships by relevant criteria. | Functional | High     | Student              | Search/filter returns matching internships.<br>Test: TC-INT-02       | Internship listings  |
| IPR-F-008 | The system shall allow students to apply for an available internship.                  | Functional | High     | Student              | Successful application is stored in the database.<br>Test: TC-INT-03 | Student profile      |
| IPR-F-009 | The system shall prevent a student from submitting multiple applications               | Functional | High     | Student              | Duplicate application is rejected.<br>Test: TC-INT-04                | Application database |

|            |                                                                                              |            |      |         |                                                                                                    |                    |
|------------|----------------------------------------------------------------------------------------------|------------|------|---------|----------------------------------------------------------------------------------------------------|--------------------|
|            | for the same internship.                                                                     |            |      |         |                                                                                                    |                    |
| IPR-F-0010 | The system shall allow students to view the current status of their internship applications. | Functional | High | Student | Student can view submitted, shortlisted, rejected or other configured statuses.<br>Test: TC-INT-05 | Application module |

## 4.4 Aptitude Test Module

### Description

The system shall provide online aptitude tests for placement preparation.

| Req ID    | Requirement                                                                 | Type       | Priority | Source / Stakeholder | Acceptance / Criteria / Test Case                                  | Dependencies      |
|-----------|-----------------------------------------------------------------------------|------------|----------|----------------------|--------------------------------------------------------------------|-------------------|
| IPR-F-011 | The system shall allow students to participate in available aptitude tests. | Functional | High     | Student              | Student can start and complete an available test. Test: TC-TEST-01 | Test module       |
| IPR-F-012 | The system shall calculate the                                              | Functional | High     | Student              | Score is calculated correctly                                      | Question database |

|           |                                                                    |            |      |         |                                                                      |          |
|-----------|--------------------------------------------------------------------|------------|------|---------|----------------------------------------------------------------------|----------|
|           | student's score automatically after test submission.               |            |      |         | based on submitted answers.<br>Test: TC-TEST-02                      |          |
| IPR-F-013 | The system shall store the student's test scores and test history. | Functional | High | Student | Completed test results are stored and displayed.<br>Test: TC-TEST-03 | Database |

## 4.5 Placement Preparation Progress

### Description

The system shall help students monitor their preparation progress.

| Req ID    | Requirement                                                                          | Type       | Priority | Source / Stakeholder | Acceptance / Criteria / Test Case                            | Dependencies              |
|-----------|--------------------------------------------------------------------------------------|------------|----------|----------------------|--------------------------------------------------------------|---------------------------|
| IPR-F-014 | The system shall allow students to record and update placement preparation progress. | Functional | Medium   | Student              | Progress updates are saved successfully.<br>Test: TC-PROG-01 | Student profile           |
| IPR-F-015 | The system shall display performance analytics based on aptitude test                | Functional | High     | Student              | Analytics are generated from available student data.         | Test and progress modules |

|  |                                   |  |  |  |                  |  |
|--|-----------------------------------|--|--|--|------------------|--|
|  | results and preparation progress. |  |  |  | Test: TC-PROG-02 |  |
|--|-----------------------------------|--|--|--|------------------|--|

## 4.6 Recruiter Module

### Description

Recruiters shall be able to manage internship opportunities and applications.

| Req ID    | Requirement                                                                         | Type       | Priority | Source/Stakeholder | Acceptance Criteria / Test Case                                              | Dependencies       |  |
|-----------|-------------------------------------------------------------------------------------|------------|----------|--------------------|------------------------------------------------------------------------------|--------------------|--|
| IPR-F-016 | The system shall allow recruiters to create, update and manage internship listings. | Functional | High     | Recruiter          | Recruiter can create and modify their internship listings. Test: TC-REC-01   | Recruiter account  |  |
| IPR-F-017 | The system shall allow recruiters to view and update internship applications.       | Functional | High     | Recruiter          | Recruiter can view applications and update candidate status. Test: TC-REC-02 | Application module |  |



## 4.7 Administration Module

| Req ID    | Requirement                                                                                       | Type       | Priority | Source/Stakeholder | Acceptance Criteria / Test Case                                          | Dependencies         |  |
|-----------|---------------------------------------------------------------------------------------------------|------------|----------|--------------------|--------------------------------------------------------------------------|----------------------|--|
| IPR-F-018 | The system shall allow administrators to manage users, internships, aptitude tests and questions. | Functional | High     | Admin              | Admin can add, update or remove permitted system data. Test: TC-ADMIN-01 | Admin authentication |  |

## 5. Non-Functional Requirements

The following requirements define the quality and operational characteristics of the system.

At least **5 NFRs** are required. This SRS contains **8 NFRs**.

| Req ID     | Requirement                                                                                | Category    | Priority | Acceptance Criteria / Measurement                                  |
|------------|--------------------------------------------------------------------------------------------|-------------|----------|--------------------------------------------------------------------|
| IPR-NF-001 | The main dashboard shall load within 3 seconds under normal network and server conditions. | Performance | High     | Response time measured using performance testing. Test: TC-PERF-01 |
| IPR-NF-002 | The system shall provide at                                                                | Reliability | High     | Availability monitored during                                      |

|            |                                                                                                    |                      |        |                                                                                       |
|------------|----------------------------------------------------------------------------------------------------|----------------------|--------|---------------------------------------------------------------------------------------|
|            | least 99% availability during normal operating periods.                                            |                      |        | testing/deployment.<br>Test: TC-REL-01                                                |
| IPR-NF-003 | The system shall support at least 500 concurrent users under the planned deployment configuration. | Scalability          | Medium | Load testing shall be performed with simulated concurrent users.<br>Test: TC-SCALE-01 |
| IPR-NF-004 | The user interface shall be responsive on desktop, tablet and mobile screen sizes.                 | Usability            | High   | UI tested on different screen sizes. Test: TC-UI-01                                   |
| IPR-NF-005 | The system shall validate user inputs before storing them in the database.                         | Security/Reliability | High   | Invalid inputs are rejected. Test: TC-SEC-01                                          |
| IPR-NF-006 | The system shall maintain consistent application, test and user records.                           | Data Integrity       | High   | Database records remain consistent after operations.<br>Test: TC-DATA-01              |
| IPR-NF-007 | The system shall be maintainable using modular frontend and backend components.                    | Maintainability      | Medium | Modules can be modified without affecting unrelated modules. Test: TC-MAIN-01         |

|            |                                                                        |           |        |                                                                |
|------------|------------------------------------------------------------------------|-----------|--------|----------------------------------------------------------------|
| IPR-NF-008 | The system shall provide clear error messages when an operation fails. | Usability | Medium | User receives understandable error feedback.<br>Test: TC-UX-01 |
|------------|------------------------------------------------------------------------|-----------|--------|----------------------------------------------------------------|

## 6. Security

### 6.1 Security Objectives

#### SO-001 – Confidentiality

The system shall protect personal, academic, resume, application and account information from unauthorized access.

#### SO-002 – Authentication and Integrity

The system shall ensure that only authenticated and authorized users can access protected functions and that important records cannot be modified by unauthorized users.

### 6.2 Security Requirements

| Req ID     | Requirement                                                                                | Type     | Priority | Acceptance Criteria / Test Case                                      |
|------------|--------------------------------------------------------------------------------------------|----------|----------|----------------------------------------------------------------------|
| IPR-SR-001 | The system shall require authentication before allowing users to access protected modules. | Security | High     | Unauthenticated users cannot access protected pages. Test: TC-SEC-01 |
| IPR-SR-002 | The system shall store user                                                                | Security | High     | Database inspection                                                  |

|            |                                                                                                                |          |      |                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------|----------|------|----------------------------------------------------------------------------|
|            | passwords using secure password hashing instead of plaintext storage.                                          |          |      | confirms passwords are not stored in plaintext. Test: TC-SEC-02            |
| IPR-SR-003 | The system shall enforce role-based authorization for Student, Recruiter and Admin accounts.                   | Security | High | Users cannot access functions outside their assigned role. Test: TC-SEC-03 |
| IPR-SR-004 | The system shall use HTTPS/TLS for secure communication between the client and server in production.           | Security | High | Network traffic is protected using HTTPS. Test: TC-SEC-04                  |
| IPR-SR-005 | The system shall validate and sanitize user input to reduce the risk of injection and malicious input attacks. | Security | High | Malicious/invalid input is rejected or safely handled. Test: TC-SEC-05     |

## 7. Quality Attributes & Acceptance Tests

### 7.1 Quality Attributes

The system shall demonstrate the following quality attributes:

#### Performance

The system should respond quickly to user requests and provide smooth navigation.

#### Reliability

User applications, test results and progress records should remain consistent and available.

## Usability

The interface should be simple enough for students and recruiters to use without extensive training.

## Security

User credentials and personal information should be protected from unauthorized access.

## Scalability

The architecture should allow the system to support increasing numbers of students, recruiters, internships and test records.

## Maintainability

The application should use modular frontend and backend components so that future changes can be made easily.

## 7.2 Acceptance Test Criteria

The project shall be considered acceptable when:

1. All high-priority functional requirements are implemented.
2. Students can register and log in successfully.
3. Students can search and apply for internships.
4. Students can view their application status.
5. Students can take aptitude tests.
6. Test scores are calculated correctly.
7. Student preparation progress and analytics are displayed.
8. Recruiters can create and manage internship listings.
9. Recruiters can view and update applications.
10. Administrators can manage users, internships and tests.
11. Security requirements are verified.
12. No critical functional or security defects remain.
13. RTM shows that all major requirements are mapped to test cases.

## 8. System Models and UML Diagrams

At least **2 UML Use-Case Diagrams** shall be prepared.

### 8.1 UML Use-Case Diagram 1 – Student Module

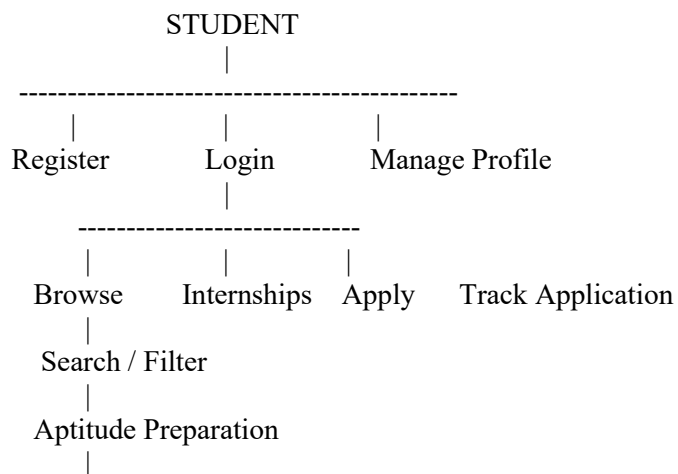
Actor

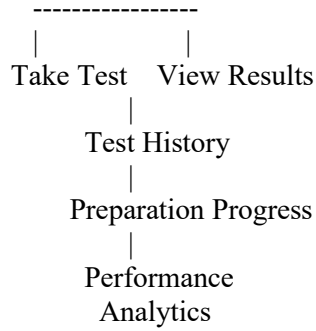
**Student**

Use Cases

- Register
- Login
- Manage Profile
- Browse Internships
- Search/Filter Internships
- View Internship Details
- Apply for Internship
- Track Application
- Take Aptitude Test
- View Test Result
- View Test History
- Update Preparation Progress
- View Performance Analytics
- Logout

Suggested relationship





## 8.2 UML Use-Case Diagram 2 – Recruiter and Admin Module

### Actors

- Recruiter
- Admin

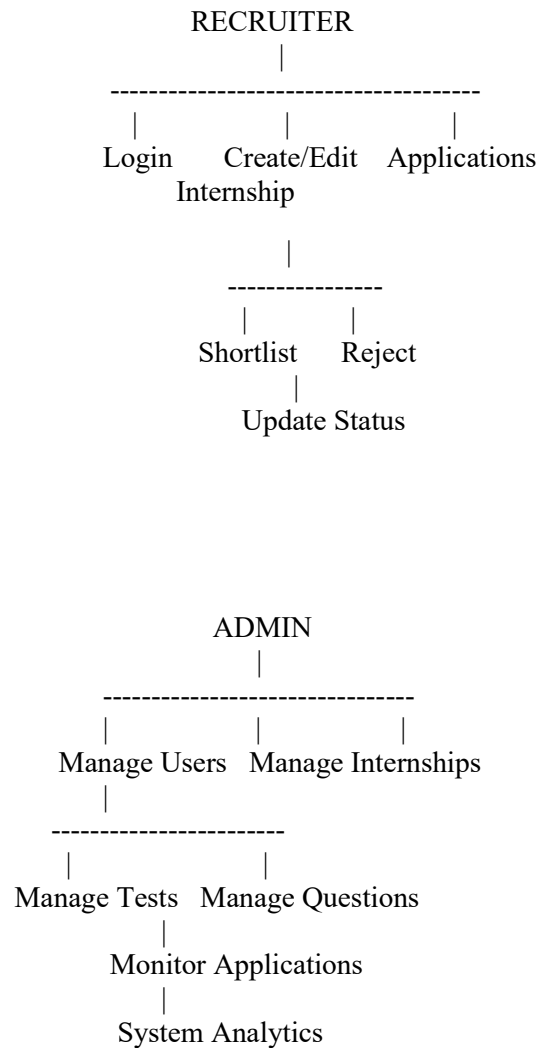
### Recruiter Use Cases

- Register/Login
- Manage Recruiter Profile
- Create Internship
- Update Internship
- Delete/Close Internship
- View Applications
- Shortlist Candidate
- Reject Candidate
- Update Application Status

### Admin Use Cases

- Login
- Manage Students
- Manage Recruiters
- Manage Internships
- Manage Aptitude Tests
- Manage Questions
- Monitor Applications
- View System Analytics

## Suggested structure



## 9. Requirements Traceability Matrix (RTM)

The RTM maps requirements to their corresponding modules and test cases.

### Status:

- N = Not Started
- P = Partially Implemented
- A = Approved/Completed



| Req ID    | Requirement Short             | Section / Design Spec | Module          | Test Case(s) | Status | Comments |
|-----------|-------------------------------|-----------------------|-----------------|--------------|--------|----------|
| IPR-F-001 | Student registration          | 4.1 / DS-AUTH-01      | Authentication  | TC-AUTH-01   | N      |          |
| IPR-F-002 | User login                    | 4.1 / DS-AUTH-02      | Authentication  | TC-AUTH-02   | N      |          |
| IPR-F-003 | Role-based access             | 4.1 / DS-AUTH-03      | Authentication  | TC-AUTH-03   | N      |          |
| IPR-F-004 | Student profile               | 4.2 / DS-PROF-01      | Student Profile | TC-PROF-01   | N      |          |
| IPR-F-005 | Academic/skill details        | 4.2 / DS-PROF-02      | Student Profile | TC-PROF-02   | N      |          |
| IPR-F-006 | Internship listings           | 4.3 / DS-INT-01       | Internship      | TC-INT-01    | N      |          |
| IPR-F-007 | Search/filter                 | 4.3 / DS-INT-02       | Internship      | TC-INT-02    | N      |          |
| IPR-F-008 | Apply internship              | 4.3 / DS-INT-03       | Application     | TC-INT-03    | N      |          |
| IPR-F-009 | Prevent duplicate application | 4.3 / DS-INT-04       | Application     | TC-INT-04    | N      |          |
| IPR-F-010 | Track application             | 4.3 / DS-INT-05       | Application     | TC-INT-05    | N      |          |
| IPR-F-011 | Take aptitude test            | 4.4 / DS-TEST-01      | Aptitude Test   | TC-TEST-01   | N      |          |
| IPR-F-012 | Calculate test score          | 4.4 / DS-TEST-02      | Aptitude Test   | TC-TEST-02   | N      |          |
| IPR-F-013 | Store test history            | 4.4 / DS-TEST-03      | Aptitude Test   | TC-TEST-03   | N      |          |
| IPR-F-014 | Preparation progress          | 4.5 / DS-PROG-01      | Progress        | TC-PROG-01   | N      |          |
| IPR-F-015 | Performance analytics         | 4.5 / DS-PROG-02      | Analytics       | TC-PROG-02   | N      |          |
| IPR-F-016 | Manage internships            | 4.6 / DS-REC-01       | Recruiter       | TC-REC-01    | N      |          |
| IPR-F-017 | Manage applications           | 4.6 / DS-REC-02       | Recruiter       | TC-REC-02    | N      |          |
| IPR-F-018 | Admin management              | 4.7 / DS-ADMIN-01     | Admin           | TC-ADMIN-01  | N      |          |

|            |                           |                 |                |             |   |  |
|------------|---------------------------|-----------------|----------------|-------------|---|--|
| IPR-NF-001 | Response time             | 5 / DS-PERF-01  | Web/API        | TC-PERF-01  | N |  |
| IPR-NF-002 | Availability              | 5 / DS-REL-01   | System         | TC-REL-01   | N |  |
| IPR-NF-003 | Concurrent users          | 5 / DS-SCALE-01 | System         | TC-SCALE-01 | N |  |
| IPR-NF-004 | Responsive UI             | 5 / DS-UI-01    | Frontend       | TC-UI-01    | N |  |
| IPR-NF-005 | Input validation          | 5 / DS-SEC-01   | Backend        | TC-SEC-01   | N |  |
| IPR-NF-006 | Data integrity            | 5 / DS-DATA-01  | Database       | TC-DATA-01  | N |  |
| IPR-NF-007 | Maintainability           | 5 / DS-MAIN-01  | All Modules    | TC-MAIN-01  | N |  |
| IPR-NF-008 | Error handling            | 5 / DS-UX-01    | Frontend/API   | TC-UX-01    | N |  |
| IPR-SR-001 | Authentication protection | 6.2 / DS-SEC-01 | Authentication | TC-SEC-01   | N |  |
| IPR-SR-002 | Password hashing          | 6.2 / DS-SEC-02 | Authentication | TC-SEC-02   | N |  |
| IPR-SR-003 | Role authorization        | 6.2 / DS-SEC-03 | Authorization  | TC-SEC-03   | N |  |
| IPR-SR-004 | HTTPS/TLS                 | 6.2 / DS-SEC-04 | Communication  | TC-SEC-04   | N |  |
| IPR-SR-005 | Input sanitization        | 6.2 / DS-SEC-05 | Backend        | TC-SEC-05   | N |  |

## 10. Summary

The **Online Internship & Placement Preparation Portal** provides an integrated platform for internship searching, application management and placement preparation.

The system supports three main user roles: **Student, Recruiter and Admin.**

Students can search and apply for internships, attend aptitude tests, track their preparation progress and view performance analytics. Recruiters can publish internship opportunities and manage applications. Administrators can manage users, internships, tests and questions.

The proposed system contains:

- **18 Functional Requirements**
- **8 Non-Functional Requirements**
- **2 Security Objectives**
- **5 Security Requirements**
- **2 UML Use-Case Diagrams**
- **Complete Requirements Traceability Matrix**

The system is planned using the **MERN stack: MongoDB, Express.js, React.js and Node.js**, with REST APIs connecting the frontend and backend.